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# Learning & Realizing Locomotion in Walking Robots

## Problem Statement

The goal is to unify control that has formal guarantees of stability and safety, and learning that allows for quick adaptation to changing environments and model uncertainties. Our focus will be on the walking robot (shown below). We are specifically focusing on obtaining the desired motion primitives via Deep Reinforcement Learning (D-RL) based techniques, and then track these motion primitives at each of the joints of the robot via Quadratic Programs (QP).

## Robot Platform

*Robot Specs. :*

- Carbon fiber chassis & legs
- Dimensions : 450 x 450 x 350 mm
- Weight : 3.0 kg
- Actuators : Servo motors (8 nos)
- Torque : 3.0 Nm (30 kg-cm)
- Onboard CPU : Raspberry Pi
- State estimation sensors :
  - Body orientation : IMU
  - Leg state : Magnetic encoders

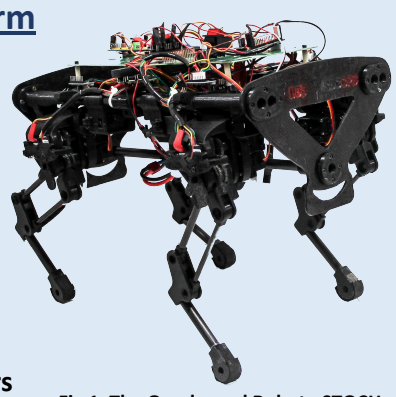

Fig 1. The Quadruped Robot : STOCH

*Challenges in Hardware :*

- High torque density requirement in joint actuators.
- High capacity & light weight mobile power supply.

## Classical Approach

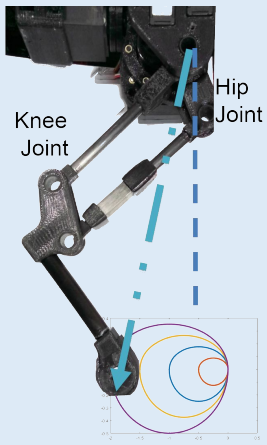


Fig 2. Controller Framework

- The walking controller is based on the Central Pattern Generators which use morphed oscillators to generate cyclic leg behaviors.
- The foot trajectories are used to obtain the desired joint angles via a standard inverse kinematics solver.
- With these joint angles, walking behaviors are generated in the robot.
- Synchronization and phase tuning of the legs results in realistic walking gaits that mimics nature.

## Fall Recovery

A dome shelled tortoise uses the inherent geometry of its shell to recover from an inverted position. Using this as the motivation, we have designed a shell for the quadruped that allows it passively roll over to an upright position all the time. The underlying principle is that the c.g. of the entire robot is never aligned with the geometric center of the robot.

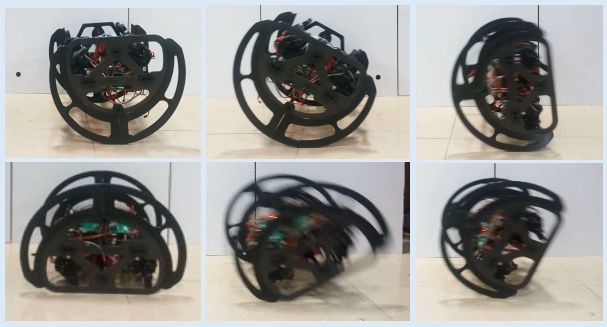

Fig 3. Frame sequence of the robot recovering from a fall

## Learning Robust Quadruped Locomotion

A Deep Reinforcement Learning (Deep-RL) agent learns locomotion behaviour from scalar rewards. The goal is to obtain a stable and faster walking behavior with better energy efficiency.

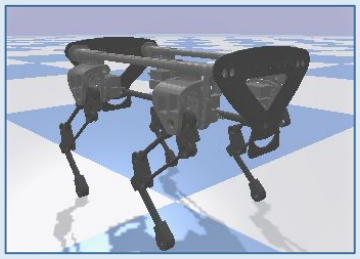

Fig 4. Robot model in Pybullet simulator

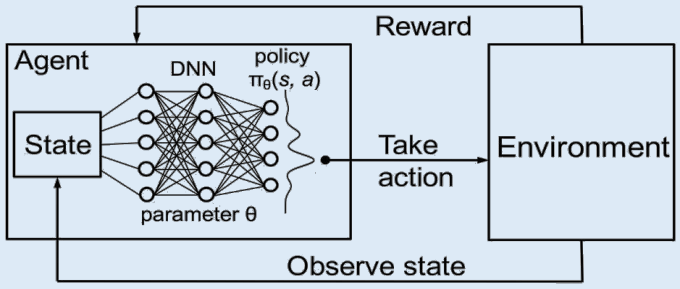


Fig 5. Reinforcement Learning framework

*Challenges*

- Transferability of the learned policy from simulation to the real physical system.
- Learning on an 8 dimensional continuous action space.
- Sampling inefficiencies of the learning algorithms.
- Running real time control of a quadruped platform with limited onboard resources, such as battery capacity, weight & computational power.

*Approach*

- Build a physics based simulation platform: PyBullet engine. Accurate motor model and hardware latencies are replicated in simulation.
- Proximal Policy Optimization (PPO) agent is utilized to explore and learn walking behaviors.
- A neural network policy learns a gaussian distribution over action which improves the exploration.
- 30 agents learn parallely, drastically reducing the training time.
- Agent learns on a variable mass distribution for better generalizability.

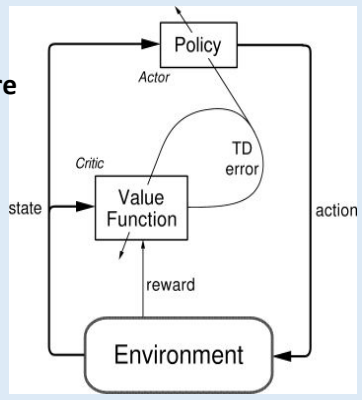


Fig 6. Actor-Critic learning framework

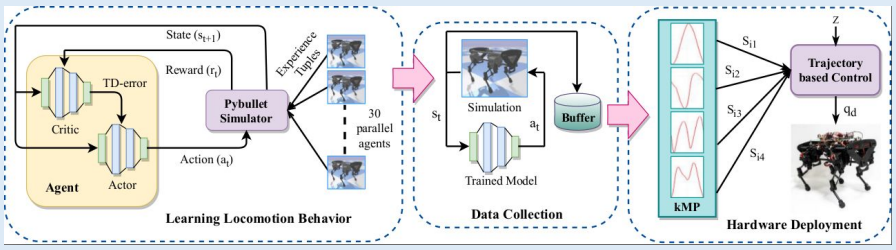


Fig 7. Framework for learning and transferability of the desired behaviour

*Progress*

- The trained RL agent achieves an average speed of 1.98 m/s within 4 hours of training.
- The robot learns and adapts the walking primitives according to the random terrain.

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